27-7-26

Experiment - 3

Design a simple network topology using cisco packet tracer

AIM:

To design and configure the simple network topology with one router, two switches and multiple PCs using cisco packet tracer.

REQUIREMENTS:

Cisco packet tracer 8.x1, Router (2811 or 1841), 2 switches (2960), 4-6 PCs Ethernet cables.

PROCEDURE:

1) Open cisco packet tracer and create new project
2) Drag 1 router, 2 switches, and 4 PC onto the workshop.
3) Connect PC0 & PC1 with switch 0 using copper straight through cable
4) connect PC2 and PC3 with switch 1 using copper straight through cable.
5) Connect switch 0 to router. via Ethernet 0/0
6) Connect switch 1 to router via ethernet 0/1

Commands

Router > enable

Router # config terminal

Router (config) # hostname main-router

mainrouter (config) # interface gi ethernet 0/1

mainrouter (config) # ip address 192.168.1.1 255.255.255.

mainrouter (config) # no shutdown

mainrouter (config-if) # exit

mainrouter (config-if) # interface gi ethernet 0/1

mainrouter (config-if) # ip address 192.168.2.1 255.255.25

mainrouter (config-if) # no shutdown

mainrouter (config-if) # end

mainrouter show ip interface brief

mainrouter # show ip route

mainrouter # copy running-config startup-config

7) Config PC0 = ip = 192.168.1.2 , mask = 255.255.255.0
GW = 192.168.1.1

8) Config PC1 = ip = 192.168.1.3, mask = 255.255.255.0
GW = 192.168.2.1

9) Config PC2 = ip = 192.168.2.2, mask = 255.255.255.0
GW = 192.168.2.1

10) Config PC3 = ip = 192.168.2.3, mask = 255.255.255.0
GW = 192.168.2.2

11) Config router int gi0/1: ip address 192.168.2.1
255.255.255.0 → no shutdown

RESULT:

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A network topology with 1 router, 2 switch, and 4 PCs was successfully created internetwork communication between the 2 subnets was verified using ping commands through cisco packet tracer.